

KW 9: The Goods Market

Ch 3, Ch 5.1, Ch 14.1-14.2 + Lecture 2

Blanchard - Macroeconomics, 7th Global Edition

Study Notes

LECTURE 2 CONTEXT: THE GOODS MARKET

Motivation: The Corona Crisis

Using the years 2020-2023, marked by the Corona crisis, we can illustrate many macroeconomic issues with real-world examples.

- Private consumption and GDP fell sharply in 2020
- At the same time, government spending increased
- Swiss data (real index, 2020 = 100): GDP and private consumption dropped ~3-4%, while government consumption rose
- Recovery was uneven: GDP recovered faster than private consumption

Motivation: The Corona Crisis

- ▶ Using the years 2020–2023, marked by the Corona crisis, we can illustrate many macroeconomic issues with real-world examples.
- ▶ Private consumption and GDP fell — at the same time, government spending increased.

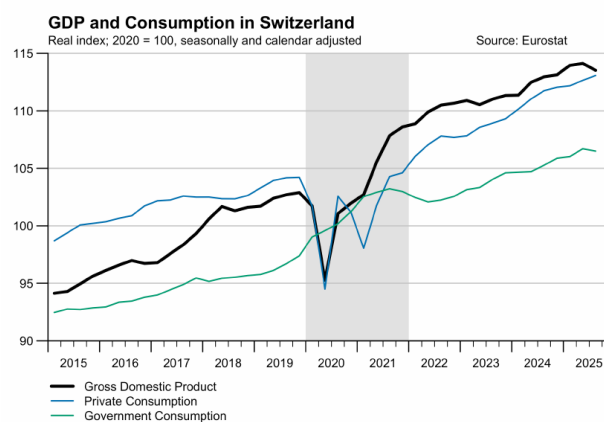


Figure: Corona Crisis - Swiss GDP, Private Consumption, and Government Consumption (real index, 2015-2025)

How Economists Analyze Economic Policy Problems

Using (mathematical) models:

- Precision of the problem, transparent assumptions, consistent analysis
- Clear indication of model assumptions and causal relationships
- Basic concept of an economic model:
 - > Mathematical functions describe causal relationships
 - > Model assumptions influence both causality and dynamics
 - > **Exogenous** (independent) and **endogenous** (dependent) variables
 - > Model parameters

- The model shows dynamics and is suitable for **comparative statics**
- Comparison of model statements with empirical analysis

Empirical Macroeconomics

- Empirical analyses require experiences and real events
- In macroeconomics there are no artificial laboratory data to test hypotheses - one must use available data based on past real situations
- Important econometric insights are based on:
 - > Time series data
 - > 'Natural experiments' (e.g. German reunification)
 - > Comparisons between countries (or states, cantons)
- Example: Did the Hartz reforms reduce German unemployment?

Learning Objectives (Lecture 2)

- The economy in the short-run
- Idea and construction of a macroeconomic model
- Components of gross domestic product (GDP) and their determinants
- Consumption theories according to Keynes and Friedman
- Derivation of the IS curve for the IS-LM model
- Literature: Blanchard Chapter 3, 5.1, 14.1 & 14.2

GDP Over Time and Course Structure

In the long run GDP follows a (hopefully rising) path. At any given time there is a GDP value on the trend line:

$$Y_{n,t} \quad (\text{natural/trend GDP at time } t)$$

- We first examine **short-term fluctuations** (-> Lectures 2 to 4)

Then we determine $Y_{n,t}$ at a given time t (-> Lectures 5 to 7)

Finally we analyze the trend of $Y_{n,t}$ (-> Lectures 10 to 11)

The Economy in the Short-Run

- In lectures 2, 3, and 4 we focus on the **short-run**:
 - > Short-run (< 5 years), medium-run (5-10 years), long-run (> 10 years)
 - > The subdivision is based on how quickly economic processes unfold
 - > Each requires a different perspective and economic analysis tools
- Economic variables behave differently in the short-run than in the long-run

- Example: Government spending is increased. Short-term GDP increase; the long-term effect is much less clear.
- In the short-run, we ignore resource constraints and analyze economic fluctuations. Prices and wages are rigid in the short run.

The IS-LM Model Preview

We use the well-known IS-LM model for short-run analysis:

- As simple as possible, but not too simple
- We essentially need: **goods market + financial market**
- We leave out for now: labor market, stocks, foreign countries, ...
- Goods market in equilibrium (IS curve): supply and demand
- Financial market in equilibrium (LM curve): supply and demand
- Goal: Understanding model, applying model, questioning model

Example: Japan VAT Increase (2019)

- Japan raised the value added tax from 8% to 10% as of October 1, 2019
- Bigger than expected hit from the sales tax rise; drop in spending
- Japan's government increased VAT knowing the short-term negative effects - the goal was to reduce the public deficit
- How is the measure to be assessed in the medium-term? -> This is what our models help answer

CHAPTER 3: THE GOODS MARKET

Year-to-year movements in economic activity are driven by the interactions among production, income, and demand. Changes in demand lead to changes in production, which lead to changes in income, which in turn lead to changes in demand.

The Economy in the Short-Term (Lecture)

With annual fluctuations in economic activity, the interrelation between production, income, and demand is central to macroeconomic understanding:

- Changes in demand lead to adjustments in production
- Adjustments in production trigger changes in income
- Changes in income cause changes in demand
- In the analysis of the goods market we therefore focus on demand for goods, production, and income - as seen in Lecture 1, all three correspond to GDP

Swiss GDP Growth Over Time (Lecture)

Expenditure-side growth contributions to nominal GDP (Source: SECO, 2000-2024): Private consumption, public consumption, investment, net exports all contribute to GDP growth. Key observation:

- **Consumption fluctuates much less than GDP** (consumption smoothing)
- Investment and net exports are the main sources of volatility

3-1 The Composition of GDP

GDP can be decomposed from the point of view of different buyers for goods. The terms 'output' and 'production' are synonymous.

The GDP can be determined in three ways: production approach, income approach, expenditure approach (see Lecture 1). Here we use the expenditure approach:

$$GDP = Y = C + I + G + NX$$

where $NX = X - IM$ (net exports). Assumption for a closed economy: $X = IM = 0$.

The Five Components of GDP

Component	Symbol	US 2014 (\$B)	% of GDP
Consumption	C	11,865	68.3%
Investment	I	2,782	16.0%
- Nonresidential		2,233	12.9%
- Residential		549	3.1%
Government spending	G	3,152	18.1%
Net exports	X-IM	-530	-3.1%
Inventory investment		77	0.4%
GDP (Y)		17,348	100.0%

Detailed Definitions

- **Consumption (C):** Goods and services purchased by consumers. Largest component at 68% of GDP.
- **Investment (I):** Also called fixed investment. Sum of nonresidential investment (firms buying plants/machines) and residential investment (people buying houses). NOT financial investment.
- **Government spending (G):** Purchases of goods and services by government. Does NOT include government transfers (Medicare, Social Security) or interest on government debt. $G = 18.1\%$ of GDP, much less than total government spending (~33%) because transfers are excluded.
- **Net exports (X - IM):** Exports minus imports. Trade surplus if $X > IM$; trade deficit if $X < IM$. U.S. had a trade deficit of 3.1% in 2014.
- **Inventory investment:** Difference between goods produced and goods sold. $\text{Production} > \text{Sales} = \text{positive inventory investment}$.

3-2 The Demand for Goods

Total demand for goods is denoted by Z:

$$Z \equiv C + I + G + X - IM$$

Three Simplifying Assumptions

- (1) All firms produce the same good (one market)
- (2) Firms supply any amount at a given price P (focus on demand; valid in short run)
- (3) Closed economy: no trade

$$X = IM = 0$$

Under closed economy assumption:

$$Z = C + I + G$$

Consumption (C)

- Consumption depends primarily on **disposable income**:

$$Y_D = Y - T$$

$$C = C(Y_D) \quad (+)$$

Linear specification:

$$C = c_0 + c_1 \cdot Y_D = c_0 + c_1(Y - T)$$

- **Marginal propensity to consume:** effect of an additional \$1 of disposable income on consumption:

$$c_1 \text{ with restriction } 0 < c_1 < 1$$

- Example: if $c_1 = 0.6$, an additional \$1 of income raises consumption by \$0.60
- **Autonomous consumption:** consumption when disposable income is zero. Must be positive (people still need to eat):

$$c_0 > 0$$

- Changes in c_0 reflect changes in consumer confidence, ease of borrowing, etc.

Components of GDP — Consumption (graphical)

► Graph:

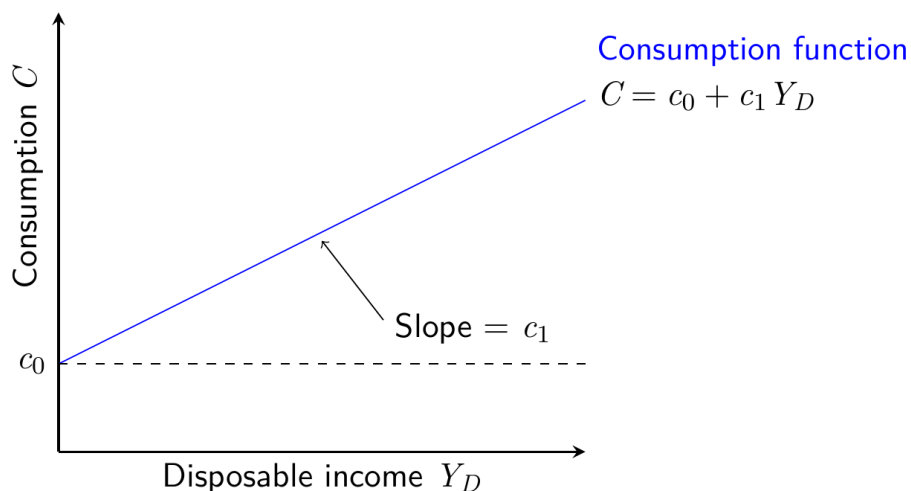


Figure: The Linear Consumption Function - $C = c_0 + c_1 * Y_D$ (slope = c_1 , intercept = c_0)

Consumption Theories (Lecture)

Two main theories of consumption:

- **Keynesian consumption function** (used in this model): Consumption is based only on current disposable income. The propensity to save/consume is postulated from aggregated data (no micro-foundation).
- **Permanent income hypothesis** (Friedman): Households with a longer-term planning horizon choose consumption to maximize an intertemporal utility function (-> consumption smoothing)
 - > Elements of the permanent income model:
 - > Lifetime income: all present and future income
 - > Utility function: consumption and leisure generate utility
 - > Intertemporal budget constraint: households cannot spend more than they earn
 - > Optimization within the constraints (-> tutorials)

- We focus on the Keynesian approach in this chapter

Keynesian Consumption: Swiss Empirical Data (Lecture)

Based on Swiss data for disposable income (Y_D) and consumption (C), the estimated function is:

$$\hat{C} = \hat{C}_0 + \hat{C}_1 Y_D = 68 + 0.8 Y_D$$

- The estimated marginal propensity to consume is approximately 0.8
- Source: Federal Statistical Office, data 1990-2024 in billion CHF

Keynesian Consumption Function in the Data

- Based on Swiss data for income (Y_D) and consumption (C) the result is an estimated function $C = \hat{C}_0 + \hat{C}_1 Y_D = 68 + 0.8 Y_D$.

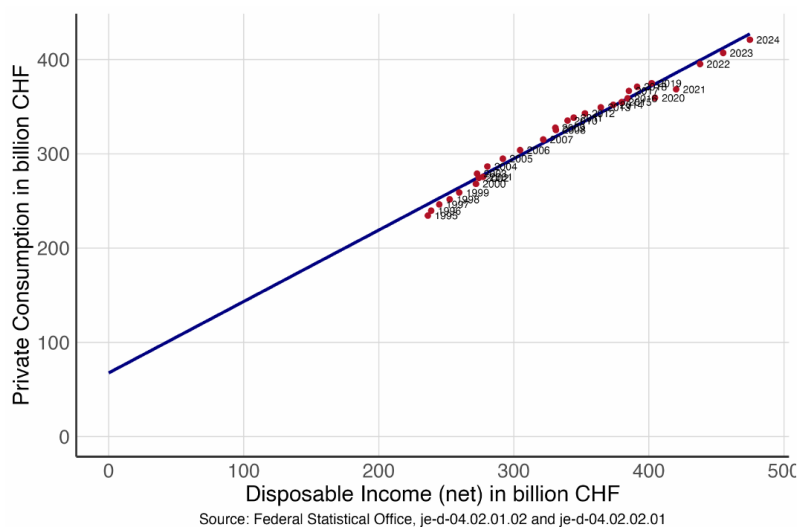


Figure: Swiss Empirical Consumption Function - $C = 68 + 0.8 * Y_D$ (scatter + regression, 1990-2024)

Critique of Keynesian Consumption (Lecture)

- Every economic model requires simplifying assumptions - it is important to critically examine them
- Fundamental critique: the function is postulated and not derived (not micro-founded)
- Important factors NOT considered: income shifts over the life cycle, consideration of future income (and taxes), savings and interest rate, wealth

Investment (I)

- Treated as **exogenous** (taken as given) in this chapter:

$$I = \bar{I}$$

- This simplification is relaxed in Chapter 5 where investment depends on output and the interest rate

Investment: Lecture Detail (Preview of Ch 5)

Three components of private investment: investment in equipment by firms, investment in construction, changes in

inventories.

- Private investment is important for two reasons:
 - > (1) Part of GDP: investment explains the level and (short-term) fluctuations in output
 - > Investment is much more volatile than consumption (-> household consumption smoothing)
- (2) Increase the stock of productive capital:

$$K_{\text{tomorrow}} = K_{\text{today}} + I - \text{Depreciation}$$

- Investment central to long-term economic growth (-> Lecture 10)
- What determines the level of private investment?
- **(1) The interest rate:** Companies have many potential investment projects with different returns-on-investment (RoI). Every project has a **net present value**:

$$NPV = CF_0 + \frac{CF_1}{1+i} + \frac{CF_2}{(1+i)^2} + \frac{CF_3}{(1+i)^3} + \dots$$

CF_0 = investment today (negative cash flow)

$CF_1, CF_2, CF_3, \dots$ = future earnings (expected)

- A lower interest rate makes more projects profitable (-> lower opportunity costs)
- **(2) The level of production:** To produce more, additional machines are necessary

Summarizing both aspects mathematically:

$$I = I \left(\underset{(+)}{Y}, \underset{(-)}{i} \right)$$

- In contrast to Blanchard, who often writes just $I(i)$, the lecture emphasizes $I(Y, i)$
- The model could be supplemented by other factors: expectations about the future (-> PMI), risk preferences
- **Endogenous variables:** depend on other variables in the model
- **Exogenous variables:** not explained in the model (taken as given)

Government Spending (G) - The State

$$\text{GDP: } Y = C + I + G + NX$$

How can the state act to influence GDP?

- Direct and indirect taxes -> T
- Transfer payments (e.g., unemployment benefit) -> T
- Government spending on goods and services -> G
- Decisions on government spending G, and the amount of taxes and transfers T, are called **fiscal policy**

- Both G and T treated as exogenous: (1) governments don't behave with simple regularity, (2) we want to analyze policy scenarios

3-3 The Determination of Equilibrium Output

Combining all components:

$$Z = c_0 + c_1(Y - T) + \bar{I} + G$$

Equilibrium condition (production = demand):

$$Y = Z$$

Three Types of Equations in Models

1. Identities (e.g., disposable income definition)
2. Behavioral equations (e.g., consumption function)
3. Equilibrium conditions (e.g., production = demand)

Substituting demand into equilibrium:

$$Y = c_0 + c_1(Y - T) + \bar{I} + G$$

Using Algebra

Solving for Y:

$$Y = \frac{1}{1 - c_1} [c_0 + \bar{I} + G - c_1 T]$$

- **Autonomous spending:** the part of demand that does NOT depend on output

$$\text{Autonomous spending} = c_0 + \bar{I} + G - c_1 T$$

- **The Multiplier:** multiplies autonomous spending. Always > 1 because $0 < c_1 < 1$:

$$\text{Multiplier} = \frac{1}{1 - c_1}$$

- Example:

$$\text{if } c_1 = 0.6, \quad \text{multiplier} = \frac{1}{0.4} = 2.5$$

- A \$1 billion increase in autonomous spending increases output by \$2.5 billion

Using a Graph: The Keynesian Cross (Figure 3-2)

The Keynesian cross diagram (lecture version):

- Y-axis: Demand Z , Production Y
- X-axis: Income Y
- **45-degree line** (Production: $Y = Y$, slope = 1)
- **ZZ curve** (demand): intercept = autonomous spending, slope = c_1 (< 1 , flatter than 45-degree line)

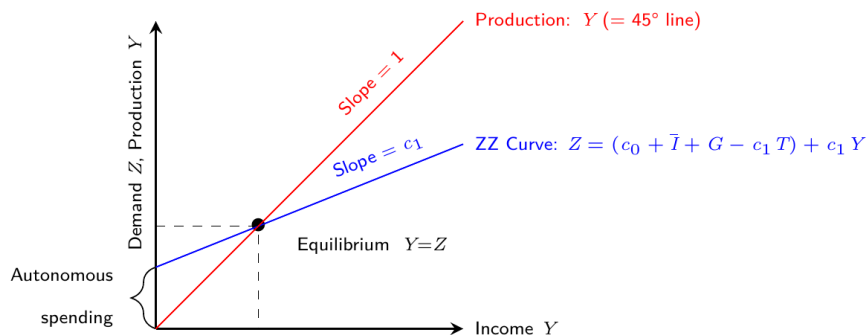
The ZZ curve equation:

$$ZZ: Z = (c_0 + \bar{I} + G - c_1 T) + c_1 Y$$

- Equilibrium at point A where ZZ crosses the 45-degree line ($Y = Z$)
- Left of A: demand $>$ production $\rightarrow$ firms increase output
- Right of A: production $>$ demand $\rightarrow$ firms decrease output
- Note: the ZZ curve is drawn for a given interest rate

Equilibrium on the Goods Market — Graphical

- Equilibrium on the goods market with $I = \bar{I}$:
 - Supply of goods (= production = income = Y) = demand for goods (= Z)
- Graph (Keynesian cross):



- Note: the ZZ curve is drawn for a given interest rate.

Figure: The Keynesian Cross - ZZ demand curve (slope c_1), 45-degree line, equilibrium at point A

Comparative Statics: Increase in G (Lecture)

How does the equilibrium change if government spending G increases?

- The demand for goods Z increases if: autonomous consumption (c_0) rises, government spending (G) rises, taxes (T) are reduced, interest rate (i) is lowered

In equilibrium, solving for Y :

$$Y = \frac{1}{1 - c_1} [c_0 + \bar{I} + G - c_1 T]$$

$$\Rightarrow \frac{1}{1-c_1} = \text{multiplier}$$

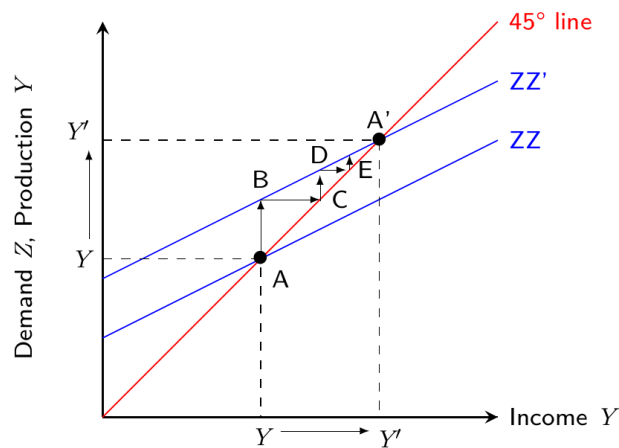
- If G increases by 1, Y increases by $1/(1-c_1) > 1$

Graphically (step by step): ZZ shifts up to ZZ' . Starting at A , demand rises to B , then production adjusts to C , then demand rises to D , then to E , converging to the new equilibrium A' .

$$\Delta Y = \frac{1}{1-c_1} \cdot \Delta G > \Delta G$$

Equilibrium on the Goods Market — Comparative Statics

- An increase in government spending G



- Note: $\Delta Y = 1/(1 - c_1) * \Delta G > \Delta G$.

Figure: Comparative Statics - Increase in G shifts ZZ up; multiplier mechanism ($A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow A'$)

How Large Is the Multiplier? (Lecture)

- Valid only for **unutilized capacities**; hence the amount is often overestimated
- Real-world examples: HSG expansion, EU Green Deal, Juncker Plan

The Multiplier Effect: Numerical Example (Lecture)

The state pays 100 CHF for road repair (with $c_1 = 0.8$):

- Road builder spends 80 CHF in a shop and saves 20 CHF
- Retailers spend $80 \times 0.8 = 64$ CHF and save 16 CHF
- After 2 rounds: cumulative increase = $100 + 80 + 64 = 244$ CHF
- After all rounds: increase in GDP by 500 CHF

$$\text{Multiplier} = \frac{1}{1-c_1} = \frac{1}{1-0.8} = \frac{1}{0.2} = 5$$

The geometric series:

$$1 + c_1 + c_1^2 + \dots + c_1^n = \frac{1}{1 - c_1} \quad (n \rightarrow \infty)$$

The Multiplier Effect (Figure 3-3)

If c_0 increases by \$1 billion:

- Round 1: Demand increases by \$1B → Production rises \$1B → Income rises \$1B
- Round 2: Consumption rises by $c_1 \times \$1B$ → Production rises by c_1 billion:

$$\Delta C_2 = c_1 \times \$1B$$

- Round 3: Consumption rises further:

$$\Delta C_3 = c_1^2 \times \$1B$$

- Total effect is a geometric series:

$$1 + c_1 + c_1^2 + \dots + c_1^n \rightarrow \frac{1}{1 - c_1} \quad (n \rightarrow \infty)$$

How Long Does Output Take to Adjust?

- In the model (no inventories, instant responses): adjustment is instantaneous
- In reality: adjustment takes time. Firms revise production quarterly, inventory adjustments occur first
- Output does not jump to Y' but increases gradually over time

Focus Box: Lehman Bankruptcy

During the 2008 crisis, consumption fell sharply despite disposable income initially not declining much. This represents a decrease in c_0 (consumer confidence collapsed). Together with falling autonomous investment (housing), the multiplier amplified these into large output declines.

3-4 Investment Equals Saving: An Alternative Equilibrium View

Private Saving

$$S = Y - T - C = -c_0 + (1 - c_1)(Y - T)$$

- **Propensity to save:** how much of an additional dollar is saved:

$$(1 - c_1) = \text{propensity to save}$$

The IS Relation

Starting from the equilibrium condition, rearranging:

$$Y = C + I + G$$

$$I = S + (T - G)$$

- In equilibrium, **investment = private saving + public saving**
- This is called the **IS relation** (Investment = Saving)
- Public saving: $T - G$. If $T > G$: budget surplus. If $T < G$: budget deficit.

Paradox of Saving (Paradox of Thrift)

If consumers try to save more (c_0 decreases):

- Equilibrium output **DECREASES**
- But total saving does **NOT** change (because $I = \bar{I}$ is fixed)
- Mechanism: more saving $\rightarrow$ less consumption $\rightarrow$ less demand $\rightarrow$ less output
 $\rightarrow$ less income $\rightarrow$ the fall in income offsets the attempt to save more
- Warning: only in the short run. In the medium/long run, higher saving can lead to higher investment and income.

3-5 Is the Government Omnipotent? A Warning

- Eq. (3.8) suggests the government can choose any output level. In reality:
 - (1) Changing G or T is slow and contentious (legislative process)
 - (2) Investment responds to output; imports leak demand abroad; exchange rates change
 - (3) Expectations matter: consumer response depends on whether tax cuts are perceived as temporary or permanent
 - (4) Side effects: too-high output can lead to increasing inflation
 - (5) Budget deficits accumulate into public debt with long-run adverse effects

CHAPTER 5, SECTION 5-1: The Goods Market and the IS Relation

Investment Now Depends on Output and Interest Rate

In Chapter 3, investment was constant. Now investment depends on two factors:

- **(1) Level of sales/output (+):** Higher sales → firms need to invest to expand capacity
- **(2) Interest rate (-):** Higher interest rate → more costly to borrow → less attractive to invest. Even with own funds, high i means it's more profitable to lend than to buy machines.

$$I = I(Y, i) \quad (+, -)$$

Determining Output

Replacing $\bar{I}$ with $I(Y, i)$ in the equilibrium condition:

$$Y = C(Y - T) + I(Y, i) + G$$

- Demand now depends on income Y (through C and I), interest rate i (through I), and fiscal policy (G, T)

Deriving the IS Curve

- For a given interest rate, the ZZ line shows demand as a function of output
- ZZ is upward sloping with slope < 1 (increase in demand $<$ increase in output)
- Equilibrium where ZZ crosses 45-degree line

Figure 5-3: The IS Curve

- An increase in i decreases investment at any level of output → ZZ shifts DOWN
- Equilibrium output DECREASES
- Plotting all (i, Y) pairs: the **IS curve is downward sloping**

IS Curve Properties

- Downward sloping: higher interest rate → lower output
- Represents all (i, Y) combinations where goods market is in equilibrium
- Shifts RIGHT with: increase in G , decrease in T , increase in consumer confidence
- Shifts LEFT with: decrease in G , increase in T , decrease in confidence

IS Curve Derivation: Two-Panel Graph (Lecture)

The IS curve is derived from the goods market equilibrium using a two-panel approach:

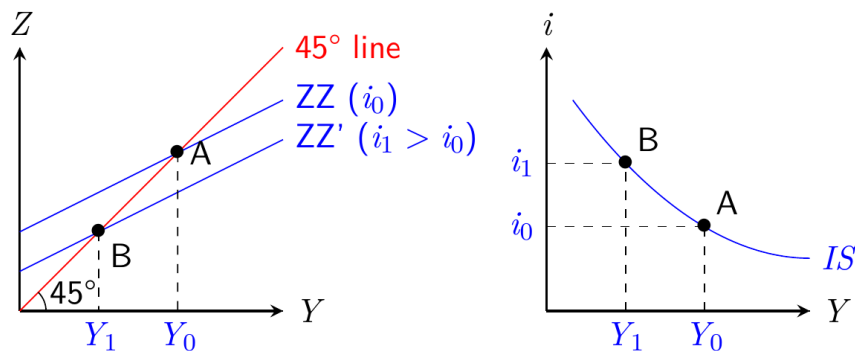
- **Left panel** (Keynesian cross): As i rises from i_0 to $i_1 > i_0$, investment falls, ZZ shifts down from $ZZ(i_0)$ to $ZZ(i_1)$, equilibrium output falls from Y_0 to Y_1
- **Right panel** (IS curve): Plot the pairs (Y_0, i_0) as point A and (Y_1, i_1) as point B, connect to get the

downward-sloping IS curve

- **Above** the IS curve: Excess supply on the goods market (production > demand)
- **Below** the IS curve: Excess demand on the goods market (demand > production)
- **On** the IS curve: Equilibrium on the goods market

Derivation of IS curve (continued)

- As interest rates rise, income falls in the goods market equilibrium. The IS curve therefore has a downward slope:



- Above the IS curve: Excess supply on the goods market
Below the IS curve: Excess demand on the goods market
On the IS curve: equilibrium on the goods market

Figure: IS Curve Derivation - Two-panel: Keynesian cross (left) and downward-sloping IS curve (right)

IS Curve: Investments and Savings (Lecture)

The name 'IS' stands for Investment = Savings. An alternative approach to the goods market equilibrium:

- Private consumer savings:

$$S = Y - T - C$$

GDP equation (excluding exports and imports):

$$Y = C + I + G$$

Rearranging:

$$S + T + C = C + I + G \implies I = S + (T - G)$$

- With trade:

$$I = S + (T - G) + (IM - X)$$

- The goods market is only in equilibrium if investment equals savings (the sum of private and state savings)
- Is S or I the driver? Theory of effective demand (Keynes) vs supply-oriented theories (Say's law, (neo-)classical)

The IS Curve: Discussion (Lecture)

The implicit IS curve:

$$Y = C(Y - T) + I(Y, i) + G \quad (\text{assumption: } X = IM = 0)$$

- The IS curve depends negatively on the interest rate: with increasing interest rates, investment demand and total production decrease
- Where does the interest rate come from? -> Financial Market, LM curve, Lecture 3
- **Movement along the IS curve:** Interest rate change -> endogenous adjustment of production volume
- **Shifts in the IS curve:** Factors that trigger a decrease (increase) in the demand for goods at a given interest rate shift the IS curve to the left (right). Example: increased government spending shifts IS right.

The Savings Paradox: Discussion (Lecture)

- Y **decreases** with increasing propensity to save (i.e., with smaller c_0 or c_1)
- The higher c_1 , the higher the multiplier
- Is saving bad? In the short run, more saving -> less consumption -> less demand -> less output
- In the medium/long run, higher saving can lead to higher investment and income

Application: Swiss Consumer Sentiment (Lecture)

Consumer sentiment in Switzerland, calculated by SECO, has continued to recover in 2025. What does this mean for the Swiss economy?

- Higher consumer sentiment -> higher c_0 (autonomous consumption) -> ZZ shifts up -> Y increases
- This is a real-world application of the multiplier mechanism

CHAPTER 14: Financial Markets and Expectations

14-1 Expected Present Discounted Values

The expected present discounted value of a sequence of future payments is the value today of this expected sequence. It allows comparing a current cost against a stream of future benefits.

Basic Concepts

- \$1 today is worth $\$(1 + i)$ next year (if lent at rate i)
- \$1 next year is worth less than \$1 today = the **present discounted value**:

$$\text{Present value of \$1 next year} = \frac{\$1}{1+i}$$

- i is the **discount rate**; the **discount factor** is:

$$\text{Discount factor} = \frac{1}{1+i}$$

- Higher interest rate $\rightarrow$ lower present value of future payments
- Example: $i = 5\%$ $\rightarrow$ \$1 next year is worth \$0.95 today
- Example: $i = 10\%$ $\rightarrow$ \$1 next year is worth \$0.91 today

The General Formula

$$\$V_t = \$z_t + \frac{1}{1+i_t} \$z_{t+1}^e + \frac{1}{(1+i_t)(1+i_{t+1}^e)} \$z_{t+2}^e + \dots$$

- Each future payment is multiplied by its discount factor
- More distant payments have smaller discount factors $\rightarrow$ lower present value
- Present value depends **positively** on future payments and **negatively** on interest rates

Special Cases

Constant Interest Rates

$$\$V_t = \$z_t + \frac{1}{1+i} \$z_{t+1}^e + \frac{1}{(1+i)^2} \$z_{t+2}^e + \dots$$

- Weights decline geometrically:

$$1, \frac{1}{1+i}, \frac{1}{(1+i)^2}, \dots$$

- $i = 10\%$: payment in 10 years $\rightarrow$ weight = 0.386; payment in 30 years $\rightarrow$ weight = 0.057

Constant Rates and Payments (n years)

$$\$V_t = \$Z \frac{1 - \left(\frac{1}{1+i}\right)^n}{1 - \frac{1}{1+i}}$$

- Example: 'Win \$1M' lottery paying \$50,000/year for 20 years at $i=6\%$: actual present value is only ~\$608,000

Perpetuity (Consol) - Payments Forever

$$\$V_t = \frac{\$Z}{i}$$

- \$10/year forever at $i=5\%$ -> worth \$200. At $i=10\%$ -> worth \$100

Zero Interest Rates

- If $i = 0$, present value = simple sum of all payments (useful approximation)

Nominal vs. Real Present Values

- **Nominal**: discount dollar payments using nominal interest rates
- **Real**: discount real payments (in terms of goods) using real interest rates
- The two approaches are equivalent:

$$\frac{\$V_t}{P_t} = V_t$$

- Use nominal for bond pricing; use real for consumption/investment decisions

14-2 Bond Prices and Bond Yields

Bond Basics

- Two key dimensions: **maturity** (time of payments) and **risk** (default risk, price risk)
- **Discount bonds** (zero-coupon): single payment at maturity (the face value)
- **Coupon bonds**: periodic coupon payments plus face value at maturity
- Coupon rate = coupon payment / face value
- Current yield = coupon payment / bond price
- **Yield to maturity**: the correct measure of return

Bond Market Vocabulary

- Government bonds (Treasury bills, Bunds) vs Corporate bonds
- Bond ratings: Moody's Aaa to C; S&P AAA to D
- Risk premium: extra rate on risky bonds vs safest bonds
- Junk bonds: high default risk bonds
- T-bills: short-term government bonds (< 1 year)
- Consols: perpetual bonds (pay forever)
- TIPS: Treasury Inflation Protected Securities (indexed to inflation)

Bond Prices as Present Values

One-year discount bond (face value \$100):

$$\$P_{1t} = \frac{\$100}{1 + i_{1t}}$$

Two-year discount bond (face value \$100):

$$\$P_{2t} = \frac{\$100}{(1 + i_{1t})(1 + i_{1,t+1}^e)}$$

Arbitrage and Bond Prices

- **Arbitrage:** if investors care only about expected returns (expectations hypothesis), expected returns must be equal across bonds

$$1 + i_{1t} = \frac{\$P_{1,t+1}^e}{\$P_{2t}}$$

- This implies: bond prices = expected present values of future payments

From Bond Prices to Bond Yields

- **Yield to maturity** for a 2-year bond: the constant annual rate making price = present value

$$\$P_{2t} = \frac{\$100}{(1 + i_{2t})^2}$$

Relationship between yield and short-term rates:

$$(1 + i_{2t})^2 = (1 + i_{1t})(1 + i_{1,t+1}^e)$$

Approximation:

$$i_{2t} \approx \frac{1}{2}(i_{1t} + i_{1,t+1}^e)$$

- The 2-year rate is approximately the average of the current and expected future 1-year rate
- Generalizes: n-year rate = average of current and expected future 1-year rates over n years

The Yield Curve (Term Structure of Interest Rates)

- A graph of yield vs. maturity for bonds observed on a given day
- **Upward-sloping:** markets expect future short rates to RISE (e.g., expected economic expansion)
- **Downward-sloping (inverted):** markets expect future short rates to FALL (e.g., expected slowdown)

Figure 14-2: U.S. Yield Curves

November 2000: Slightly downward sloping

- 3-month rate: 6.2%, 30-year rate: 5.8%
- Markets expected rates to decrease slightly (economy slowing)

June 2001: Steeply upward sloping

- 3-month rate: 3.5%, 30-year rate: 5.7%
- Fed had cut rates sharply; markets expected recovery and rate rises

SUMMARY OF KEY FORMULAS

$$(3.2) \quad C = c_0 + c_1 Y_D \quad \text{Linear consumption function}$$

$$(3.3) \quad C = c_0 + c_1(Y - T) \quad \text{Consumption (income and taxes)}$$

$$(3.8) \quad Y = \frac{1}{1-c_1} [c_0 + \bar{I} + G - c_1 T] \quad \text{Equilibrium output}$$

$$(3.10) \quad I = S + (T - G) \quad \text{IS relation}$$

$$(3.11) \quad S = -c_0 + (1 - c_1)(Y - T) \quad \text{Private saving}$$

$$(5.1) \quad I = I(Y, i) \quad (+, -) \quad \text{Investment function}$$

$$(5.2) \quad Y = C(Y - T) + I(Y, i) + G \quad \text{Goods mkt equilibrium}$$

$$(14.1) \quad V = z + \frac{z^e}{1+i} + \dots \quad \text{Present discounted value}$$

$$(14.4) \quad \$P_{1t} = \frac{100}{1+i_{1t}} \quad \text{1-year bond price}$$

$$(14.5) \quad \$P_{2t} = \frac{100}{(1+i_{1t})(1+i_{1,t+1}^e)} \quad \text{2-year bond price}$$

$$(14.12) \quad i_{2t} \approx \frac{1}{2}(i_{1t} + i_{1,t+1}^e) \quad \text{Yield approximation}$$

$$V = \frac{z}{i} \quad \text{Perpetuity value}$$

$$\frac{1}{1 - c_1} \quad \text{The Multiplier}$$